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ORIGINAL PAPER

The role of management control systems for digital transformation success

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Abstract
The aim of this paper is to gain a better understanding of the role played by management control systems (MCSs) in digital transformation success (DTS) within organizations. We draw on Simons' (Levers of control, Harvard Business School Press, Boston, 1995) levers of control to investigate, first, whether the interactive use of MCSs and belief systems fosters the development of digital dynamic capabilities (DDCs) and, second, whether they can act as moderators to increase DTS. On the basis of survey data collected from a sample of 415 managers, the study employs structural equation modelling with EQS to assess the proposed relationships. Our findings partially support the hypotheses regarding the interactive use of MCSs and belief systems. While we observed a positive direct association with DDCs, the moderating effects are not significant for interactive MCSs and are negative for belief systems. Furthermore, this study encourages managers to recognize the potential of alternative uses of MCSs beyond those hypothesized, within the context of digital transformation. This study contributes to the literature by providing new insights into the role of MCSs in DTS and by highlighting the fundamental function of DDCs as mediating variable in the relationship between MCSs and DTS.

Keywords Belief systems · Digital dynamic capabilities · Digital transformation · Interactive use · Management control systems

JEL Classification M41

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1 Introduction

Digital transformation significantly changes the way in which organizations operate in the digital era. Technologies such as cloud computing, big data, blockchain, analytics, artificial intelligence and machine learning are being integrated deeply into daily business, thus opening new possibilities and transforming the source of competitive advantage for all kinds of organizations. This transformation is a process that requires particular digital capabilities and complex decisions on which management control systems (MCSs) can provide crucial support. This research aims to analyse the role of MCSs from the lens of the specific capabilities required for digital transformation: digital dynamic capabilities (DDCs).

Following Verhoef et al. (2021), who considered digital transformation to be a change in how a firm employs digital technologies to develop digital business models that help create and appropriate more value for the firm, key elements arise that go beyond traditional innovation. Digital transformation (1) entails a strategy redefinition (Ubiparipović et al. 2023); (2) is characterized by a wide scope for the organization, which goes beyond the challenges associated with creating a new product, service or process (Rogers 2016; Warner and Wäger 2019); and (3) occurs at an accelerated rate (Gong and Ribiere 2021).

To operate in this context, organizations need specific capabilities to face rapidly changing scenarios. Dynamic capabilities have been widely considered key to the integration, construction and reconfiguration of internal and external competences to address rapidly changing environments (Teece et al. 1997), such as those affected by digital transformation. Warner and Wäger (2019) adapted the framework developed by Teece et al. (1997) to the digital world, arguing that the ubiquity of new digital technologies is changing the very nature and purpose of company competences. These authors suggested that three DDCs, namely, digital sensing, digital seizing and digital transforming, are essential for succeeding in the ongoing digital journey. The question of how to foster DDCs to address effective digital transformation remains an issue not only for researchers (Sousa-Zomer et al. 2020) but also for practitioners.

Under a digital transformation process, decision makers are involved in complex and disrupting processes and must be agile in adapting or refining business models that are consistent with their strategic objectives. The overall purpose of MCSs is to influence individual behaviour and decision-making in ways that help achieve organizational objectives (Ferreira and Otley 2009; Franco-Santos and Otley 2018). Simons' (1995) four levers of control—diagnostic and interactive uses and belief and boundary systems—have been widely applied to investigate the ways in which organizations leverage their MCSs to implement their business strategies. Their effectiveness may depend not only on the intrinsic characteristics of the MCSs but also on the type of capabilities—which, in the context of this study, are digital (DDCs)—to be fostered to properly implement a company's strategy. Moreover, the development of DDCs alone does not guarantee the success of the overall digital transformation process. Instead, success should be assessed through managers' perceptions of various strategic outcomes commonly associated with such initiatives. In this study, four key dimensions are considered to define the perceived success of digital transformation (DTS): the enhancement of competitive advantage, improvement of organizational

image, increase in company value, and the extent to which expectations tied to digital initiatives are met. Indeed, recent academic analysis highlights that more than 80 % of digital transformations fail to achieve their intended goals (Oludapo et al. 2024). While some companies have faced many difficulties with respect to the task of interpreting digital threats and opportunities (for example, Nokia and its smartphone strategy in the early twenty-first century), other companies have been able to integrate digital technology into their strategies as a source of competitive advantage (such as Apple, Netflix, and many others), thereby consolidating their growth and position in the market.

While there is a line of research concerning how technologies can influence accounting (Bhimani 2020) and, more specifically, how they can influence management control (Fähndrich 2023), to the best of our knowledge, the ways in which different MCSs can foster or inhibit digital transformation have not yet been studied. The objective of this paper is twofold. The first is to identify the specific MCS lever that can influence the development of DDCs. We argue that owing to the specificity of digital capabilities, they better match the interactive use of MCSs and belief systems, which are therefore the key levers of control on which managers should focus to positively influence them. The second objective is to investigate whether these specific levers of control can act as moderators in the relationship between DDCs and DTS.

We collect survey data from a sample of 415 Spanish and Colombian organizations. Our results partially confirm the hypotheses regarding the interactive use of MCSs and belief systems and reveal outcomes for diagnostic and boundary systems in this disruptive context. This study provides insights into which specific type of MCS has a positive and direct association with the promotion of DDCs, on the one hand, and how the moderating role of MCSs influences the relationship between DDCs and DTS, on the other hand. In both cases, the findings reveal results that enhance our understanding of the factors influencing digital transformation from a management control perspective and highlight the varying effects depending on the specific MCS lever.

This study makes several contributions. First, it contributes to the management accounting literature by providing empirical evidence concerning the ways in which MCSs positively impact organizations' digital transformation processes. Second, our findings enrich our understanding of the antecedents of DDCs by identifying managerial control practices that can support digital capabilities, a topic that has hitherto not been explored in the literature, except via limited theoretical approaches. Third, we contribute to the digital transformation literature by verifying the relationship between the degree of development of DDCs and managers' perceptions of success in digital initiatives. This contribution is important because a certain level of DDCs might (or may not) be translated into what managers consider a successful process.

Following this introduction, the theoretical background of this research is provided, and hypotheses are formulated. The methodological approach and the sample are subsequently presented, followed by the findings. Finally, a discussion and related conclusions provide a summary of this research.

2 Theoretical background

2.1 Digital transformation

Digital transformation has been identified as the changes that digital technologies can cause in products, organizational structures, and the automation of processes and business models (Cozzolino et al. 2018; Hess et al. 2016). Recently, Verhoef et al. (2021) identified three stages of digital transformation: digitization, digitalization, and digital transformation. Digitization refers to the process of encoding analogue information into a digital format (i.e., into zeros and ones); digitalization refers to the ways in which information technologies (IT) or digital technologies can be used to alter existing business processes (Li et al. 2016); and digital transformation represents the most pervasive phase, which entails a company-wide change that leads to the development of business models that may be new to the focal firm or industry.

Digital transformation and innovation processes may appear to be similar concepts, but there are key differences between them. While any transformation is an innovation, not every innovation is a transformation. A transformation must include a high degree of change, i.e., it must be a fundamental and intense modification leading to enhanced outcomes. Therefore, three essential attributes must be present in a digital innovation to merit the use of the term digital transformation: (1) strategy redefinition; (2) wide scope; and (3) rapid execution.

(1) Strategy redefinition

Although technology is widely recognized as a key capability for business innovation (OECD/Eurostat 2018), digital transformation is not defined solely in terms of technologies, processes or business models considered in isolation. Indeed, there is considerable agreement that "the improvement of competitive positioning is less dependent on the technology being adopted and more by the applied strategic approach to implementation" (Ubiparipović et al. 2023, p. 34). Several approaches to the phenomenon of digital transformation can enrich and help define concepts such as digital maturity (North et al. 2019) and technological disruption (Cozzolino et al. 2018; Vial 2019) and can be used to consolidate the idea of an ongoing and necessary understanding of the relationship between digital transformation and strategy (Warner and Wäger 2019). North et al. (2019) introduced the concept of digital maturity, explaining digital transformation not only as an issue pertaining to technology adoption but also as a journey that requires changes in mindsets and leadership practices. Vial (2019) noted that digital transformation reflects the complexity of the environment and is a process in which digital technologies create disruptions (not only in organizations but also in individuals and societies), thereby triggering strategic responses to alter value creation paths and manage structural changes and organizational barriers.

(2) Wide scope

In an organizational context, digital transformation refers to a fundamental alteration leading to the emergence of a whole new form, function, or structure due to the adoption of digital technologies (Gong and Ribiere 2021). Digital transformation involves data-driven processes, digital ecosystem cooperation and interactions with new marketplace platforms (Cennamo et al. 2020) that not only transform the com-

pany as a whole but also open new opportunities for interaction with the surrounding environment.

Both strategy redefinition and wide-scope characteristics can be better understood through the examination of examples from well-known companies such as Netflix, which was initially a distributor of DVDs by mail and implemented a strategical and wide-scope transformation based on digital technologies to become a leading provider of streaming content and, later, a provider of its own content on the basis of artificial intelligence, data analytics and cloud computing. Similarly, Amazon, which started as an online bookstore, evolved into a global tech giant by using data analytics and machine learning to provide personalized product recommendations. Additionally, digital transformation is not limited to large corporations or specific industries. Even in traditional sectors such as small agricultural holdings, there are initiatives that reflect this shift. For example, some agri-tech companies are integrating technologies such as the Internet of the things (IoT), big data, analytics, and cloud computing to optimize irrigation and improve weather forecasting. This broad adoption of digital tools is driving a shift towards more data-driven decision-making, reducing risks and operating costs while enhancing competitive advantage.

(3) Rapid execution

Digital transformation involves a series of rapid and discontinuous changes that are enabled by the innovative use of digital technologies alongside the strategic leveraging of key resources and capabilities to fundamentally alter the characteristics and nature of an entity (Gong and Ribiere 2021; Mithas et al. 2013; Vial 2019). In the context of strategy implementation, the perspectives of time and speed are important in several ways. First, technology grants access to vast data volumes, necessitating efficient processing for expedited decision-making. Second, it enhances opportunities for streamlining and synchronizing both internal and external processes. Finally, technology underscores the potential for developing capabilities to swiftly design, manage, and adapt to emerging network structures (Bharadwaj et al. 2013; Ellström et al. 2022).

Managers must leverage appropriate digital resources to create differential value (Bharadwaj et al. 2013) in their firms, thereby fostering the conditions needed to overcome barriers that can prevent the development of the necessary digital capabilities to face the challenges associated with digital transformation. To accomplish this goal, they must encourage their organizations to develop digital dynamic capabilities that are specifically adapted to the rapidly changing environment and provide opportunities for DTS (Dalenogare et al. 2018; Westerman et al. 2012).

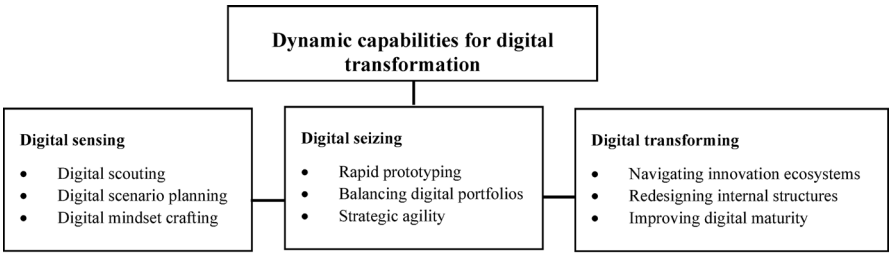
2.2 Digital dynamic capabilities

Dynamic capabilities consist of specific organizational processes such as product development, alliancing, and strategic decision-making that create value for firms within dynamic markets by manipulating resources into new value-creating strategies (Eisenhardt and Martin 2000, p. 1106). Dynamic capabilities imply the ability to renew, augment and adapt a core competence over time (Teece et al. 1997) as well as the capacity to modify or create organizational configurations to obtain a competitive advantage and improve viability (Eisenhardt and Martin 2000; Helfat 1997;

Teece and Pisano 1994; Teece et al. 1997; Winter 2003; Zollo and Winter 2002). The construct of dynamic capabilities was originally designed to answer the question of how firms can achieve and maintain competitive advantage in contexts featuring rapid technological change (Teece et al. 1997). Teece (2007) disaggregated dynamic capabilities into three main categories: the capacity to sense and shape opportunities and threats; the capacity to seize opportunities; and the capacity to maintain competitiveness by enhancing, combining, protecting and, when necessary, reconfiguring the business enterprise's intangible and tangible assets.

The dynamic capabilities framework is widely recognised as a powerful lens for examining digital transformation (North et al. 2019; Vial 2019; Warner and Wäger 2019). North et al. (2019) applied this framework to the digital context, proposing four categories to foster digital maturity: sensing digital growth potential, developing a digitally growth strategy and mindset, and seizing and managing resources to support digital transformation. Additionally, Warner and Wäger (2019) identified nine digital subcapabilities forming the basis of a DDCs framework. Their framework goes beyond the adoption of specific technologies (e.g., Gupta et al. 2020; Mikalef and Pateli 2016) and addresses the limitations of other approaches that are either non-operationalized (e.g., Vial 2019; Matarazzo et al. 2021; Canhoto et al. 2021; Magistretti et al. 2021; Ellström et al. 2022), sector-specific (e.g., Shen et al. 2022; Janssen et al. 2016; Pérez-Rave et al. 2023; North et al. 2019; Scuotto et al. 2021), or focus on isolated digital capabilities (e.g., Chae et al. 2018; Li and Chan 2019; Mikalef and Pateli 2016). This study incorporates Warner and Wäger's (2019) terminology for the three main DDCs categories: digital sensing, digital seizing and digital transforming (see Fig. 1).

Digital sensing refers to the ability to employ digital scouting and digital scenario planning to scan the external environment for new or unexpected trends (Warner and Wäger 2019). The same authors reported that the development of these capabilities is contingent on digital mindset crafting, and they emphasized the claim that companies must develop strong digitally oriented cultures. Next, digital seizing refers to the ability to implement digitalization initiatives to seize perceived opportunities (North et al. 2019). Warner and Wäger (2019) argued that agility is critical for seizing the latest trends and avoiding potential existential threats; they also claimed that this capability requires understanding the importance of rapid prototyping, providing opportunities to accelerate digital transformation, and learning to balance digital portfolios with



Adapted from Warner and Wäger (2019).

Fig. 1 Digital dynamic capabilities

the aim of being able to scale up or scale down certain business model innovations. Finally, digital transforming focuses on redesigning internal structures and establishing leadership teams as well as adopting strategies and business models with a digital focus (Warner and Wäger 2019). The strategic renewal associated with a digital transformation requires the ability to modify the organization's resource base (Agarwal and Helfat 2009). Digital transforming capability requires the ability to navigate innovation ecosystems and carry out other initiatives, such as investments and technology-based acquisitions, to improve firms' digital intensity. By improving their digital intensity, firms can increase their digital maturity (Westerman et al. 2012), maintain appropriate value and sustain digital transformation initiatives over time (Sousa-Zomer et al. 2020). Finally, to consolidate their digital transforming capabilities, companies can improve the digital maturity of the workforce and involve digital natives in the capability-building process (Warner and Wäger 2019).

Dynamic capabilities promote the process of alignment between business strategy and IT (Yeow et al. 2017) and are essential for creating, implementing, and transforming business models to remain relevant in the emerging digital economy (Karimi and Walter 2015, 2016; Teece 2018; Teece and Linden 2017; Velu 2017). Companies engaging in digital transformation processes must explore their DDCs intensively, which has been shown to be imperative for ensuring the survival of organizations in the digital age (Warner and Wäger 2019). Our approach stipulates that MCSs can both foster DDCs and improve the positive effect of DDCs on managers' perceptions of DTS.

2.3 Management control systems

Simons' (1995) levers of control (LoC) framework is well established in the management control literature, with decades of empirical application and broad scholarly adoption (e.g., Bisbe and Otley 2004; Ferreira 2009; Lopez-Valeiras et al. 2016). This study adopts the LoC framework not only because of its strong theoretical and empirical grounding but also because of its particular suitability for analysing digital transformation processes, which are inherently strategic. The central role of strategy in Simons' framework (Tessier and Otley 2012), along with its alignment with performance enhancement through capability development (Bellora-Bienengräeber et al. 2023), reinforces its relevance for the proposed research model. While alternative perspectives exist, Simons' framework represents the most robust and contextually appropriate choice for this study.

Simons defined MCSs as formal, information-based routines and procedures that managers use to maintain or alter patterns in organizational activities (Simons 1995, p. 5) and identified four levers of control: diagnostic and interactive uses and belief and boundary systems. Owing to its particular focus on strategic control, Simons' framework is useful for the purposes of this study, as it fits with the definition of digital transformation by focusing not only on the implementation of technology but also on the evolution of a company's strategy (Matt et al. 2015; Rogers 2016). Furthermore, MCSs can provide information to direct and develop the coordination, learning, and reconfiguration of processes, which are intrinsic to Teece et al.'s (1997) dynamic capabilities theory.

Boundary systems are used by top managers to define explicit limits and rules and are typically formulated in negative terms or used as minimum standards with a focus on the risks that must be avoided (Simons 1994). They usually comprise codes of conduct that outline unethical behaviours. By communicating those limits, boundary processes aim to prevent employees from wasting the organization's resources (Mundy 2010). Conversely, belief systems are used by top managers to define, communicate, and reinforce the basic values, purpose, and direction of the organization (Simons 1994). Beliefs and values are incorporated into MCSs to secure commitment to goals and to inspire employees in their search for opportunities and solutions (Marginson 2002). The design of belief systems is influenced by the company's core values and takes the form of credos or mission statements that define the organization's purpose and impart what employees should do and how they should act. Additionally, employees who demonstrate these values through their work can be recognized and rewarded.

Nevertheless, according to Simons (1994, 1995), the effectiveness of MCSs depends not only on the presence and the technical design of certain mechanisms of control but also on the ways in which managers use them. Thus, the diagnostic use of MCSs is employed to monitor organizational outcomes and provide feedback to correct deviations from preset standards (Simons 1994). The diagnostic use of MCSs is based on programmed cybernetic processes and management by exception and is used to establish guidelines for corrective action (Bisbe and Malagueño 2015; Widener 2007). Diagnostic use of control systems refers to the way performance information is used to monitor outcomes against predefined targets. Tools such as performance scorecards, project monitoring systems, or HR key performance indicators can be used diagnostically when they are employed to track deviations from goals and trigger corrective actions. These systems typically produce periodic performance reports that are compared with established targets. Finally, MCSs are interactively employed by top managers to become involved regularly and personally in the decision-making processes of subordinates by focusing organizational attention on strategic uncertainties (Bisbe and Malagueño 2015). These systems are described as incorporating frequent and forwards-looking dialogue among managers (Widener 2007), who use the systems to signal organizational priorities and stimulate the emergence of new strategies (Bisbe and Otley 2004; Naranjo-Gil and Hartmann 2007; Simons 1995). The interactive use of MCSs can be implemented, for example, through regular strategic review meetings where the leaders of each department present the progress and challenges of their key projects.

According to Healy et al. (2018) and Davila et al. (2009), Simons' (1995) definition of MCSs is consistent with the concept of dynamic capabilities, and these authors suggested that the levers of control model (Simons 1995; Simons et al. 2000) leads to a paradigm shift in terms of the extant literature on control, offering a way to create rather than limit or eliminate the conditions necessary for innovative activity to occur. However, in the disruptive context of digital transformation, in which organizations must overcome important barriers that may limit the development of the required DDCs, not every MCS lever seems to be equally appropriate. According to Henri (2006a), the diagnostic use of MCSs may not represent an adequate means of fostering capabilities related to market orientation, entrepreneurship or innova-

tiveness, as DDCs are. Similarly, boundary systems, which are usually expressed as limitations, probably do not help overcome the abovementioned barriers or resistance to change; therefore, they are not expected to play any paramount role in the digital transformation process.

In contrast, a reliance on belief systems and the interactive use of MCSs suggest that the organization actively scans its environment, searches for new strategic opportunities, and encourages experimentation and learning to overcome strategic uncertainties and risks, which are key elements of the digital transformation process (Aaltola 2018; Albertini 2019; Bisbe and Malagueño 2015; Mundy 2010; Maier and Branzei 2014; Widener 2007). In addition, owing to the high speed of execution of digital transformation, MCSs should be designed to feature a lower level of formalization and to focus more on soft skills such as creativity, knowledge sharing, commitment and flexibility. Leveraging DDCs requires cross-functional teams, fast decision-making and executive support as well as both informal and formal scouting networks, which organizations must incorporate into their strategic planning processes (Bisbe and Malagueño 2015). These requirements are better aligned with the interactive use of MCSs. Finally, belief systems can allow employees to engage in spontaneous and responsive actions and adapt to rapidly changing business conditions, which is consistent with the values of flexibility (Heinicke et al. 2016; Henri 2006b), and serve to inspire, guide, and motivate behaviour through the use of broad value-laden concepts (Mundy 2010; Simons 2000). This reasoning justifies our focus on the interactive use of MCSs and belief systems.

3 Hypotheses development

3.1 Interactive use of MCSs and belief systems to foster DDCs

As previously stated, DDCs, which are split into three main categories, play a crucial role in the digital transformation process. The interactive use of MCSs is expected to foster them. First, digital sensing can help organize, filter and spread the data obtained in the digital scouting process, generating valuable information and facilitating its flow vertically throughout the organization (Widener 2007). This type of use is considered effective for firms facing technological and environmental uncertainty (Bisbe and Otley 2004) and can be useful for planning different digital scenarios. Through participation and discussion with managers, the interactive use of MCSs can also motivate employees to adopt a digital mindset that favours the development of ideas (Mundy 2010). Second, regarding digital seizing capability, interactive MCSs allow managers to involve themselves in resource reallocation. They help properly seize digital initiatives that can take the form of digital prototypes, facilitating the translation of innovations into strategy, stimulating dialogue, and supporting the emergence of new strategies (Henri 2006a). The interactive use of MCSs introduces criteria for the management team's selection of digital portfolios, thereby providing support for answering questions related to the timing, risks, and economic consequences of alternative actions and allocating funds and assets nimbly (Müller-Stewens et al. 2020). Third, referring to transforming capability, the interactive use of

MCSs can help create an environment in which the individuals involved in making a decision feel free to offer their honest opinions and are able to consider objective data to mitigate the effects of closed thinking (Teece 2007). This trust context can favour the redesign of internal structures when necessary. Moreover, the interactive use of MCSs helps generate a two-way process of knowledge transfer that leads to work by using a collaborative approach, which is required for successfully integrating external and internal knowledge (Heinicke et al. 2016). This internal/external balance can be particularly suitable when navigating innovative digital ecosystems. Moreover, by fostering organizational dialogue and debate and encouraging information exchange, the interactive use of MCSs contributes to knowledge dissemination, information distribution and communication, which are key for enhancing companies' digital maturity (Malina and Selto 2001; Simons 1995). Accordingly, the following hypothesis is proposed:

H.1. *The interactive use of MCSs is positively associated with DDCs.*

Similarly, belief systems can favour the development of DDCs, enabling a strategic vision (Kruis et al. 2016) to be shared throughout the entire organization. First, with respect to sensing capability, belief systems can inspire and consolidate a mindset crafted to a digitally oriented organization that focuses on the need to search for digital opportunities and solutions (Marginson 2002; Widener 2007). Second, belief systems can impact seizing capability, boosting "secure" experimentation with new digital prototypes, which can help discover new value chains to create radical business model innovations (Autio et al. 2018). Belief systems play a crucial role in managers' ability to bend or break existing structures to introduce managerial innovations (Janka et al. 2020), enabling them to balance digital portfolios with existing product-based business models. They also provide guidance related to legitimacy and ways of stimulating autonomous initiatives (Bisbe and Otley 2004; Davila et al. 2009; Henri 2006a), which can facilitate the strategic agility that new business models need to exploit technological and market opportunities quickly (Warner and Wäger 2019). Increased flexibility results from the establishment of a belief system that communicates the importance of quick response to strategic demands. Third, belief systems support transforming capability through the encouragement of the ability to navigate innovation ecosystems, establish relationships with strategic agents and offer guidelines for focusing on the interests of the organization. Belief systems can foster values such as decentralization, autonomy and empowerment, which enable managers to lead with conflict resolution and effective decision-making on the basis of the organization's core values (Henri 2006b). These values facilitate, when necessary, the redesign of internal structures and the proposal of a strategic story (Aaltola 2018) that is capable of inspiring and motivating employees (Simons 1995) to favour the creation of digitally mature oriented teams. Accordingly, the following hypothesis is proposed:

H.2. *Belief systems are positively associated with DDCs.*

3.2 Effect of DDCs on company DTS

Enhancing DDCs lacks justification without an examination of their capacity to effectively underpin the success of the digital transformation process. However, assessing this success as an outcome of this process is a challenging task, mainly due to the novelty and scope of the digital transformation phenomenon (Ghobakhloo and Iranmanesh 2021). Typically, the literature has addressed the digitalization performance associated with a simple project, product or specific technology implemented; in contrast, this paper conceptualizes DTS at a more global and strategic level to investigate the achievement of company objectives resulting from digital transformation implementation. DTS, by definition, is multifaceted and may include a combination of factors tailored to company-specific circumstances and objectives. However, some recurring objectives throughout the digital transformation process are to achieve and maintain competitive advantage within the market (Medeiros and Maçada 2022; Teece et al. 1997); cultivate a positive corporate image among customers, business partners, and stakeholders (Cichosz et al. 2020); and preserve or augment the overall value of the organization for shareholders and employees (Ribeiro-Navarrete et al. 2021). These three elements collectively address the diverse array of those concerned both within and outside the organization. Additionally, and broadly, companies are keen on evaluating whether expectations, which are distinct from the three previously mentioned elements but linked with the digital transformation process, have been fulfilled. Consequently, assessing whether any other objectives are achieved as a result of engaging in a digital transformation process has also been considered. Previous studies have found only limited evidence regarding the impact of DDCs—a very new framework (2019)—on DTS,[1] and to the best of our knowledge, these relationships have not yet been tested. While Warner and Wäger (2019) posited the potential for a positive relationship between DDCs and DTS, their analysis was primarily qualitative in nature. In contrast, our study seeks to provide empirical validation of this relationship. We postulate that DDCs—digital sensing, digital seizing, and digital transforming—play a crucial role in facilitating technological change within organizations, enabling them to successfully capitalize on opportunities and achieve their objectives. Specifically, sensing capabilities enable organizations to identify emerging digital opportunities, which can enhance strategic positioning and contribute to perceived increases in competitive advantage and the organization's image. Seizing capabilities allow firms to mobilize resources and align operations to capture value from digital initiatives, thus potentially increasing company value. Finally, transforming capabilities support the continuous reconfiguration of processes and structures, which is essential for meeting the expectations associated with digital transformation initiatives. Consequently, we propose the following hypothesis:

H.3. DDCs are positively associated with DTS.

[1] We focused exclusively on the direction from DDCs to DTS, while acknowledging the potential for reciprocal reinforcement between the two constructs. However, the absence of strong theoretical grounding for considering DTS as a precursor to DDC development suggests that the risk of reverse causality is limited.

3.3 Moderating role of the interactive use of MCSs and belief systems in the relationship between DDCs and DTS

While it is hypothesized that the development of DDCs is positively associated with a successful outcome in digital transformation processes, this relationship may be either enhanced or restricted by the different uses of MCSs. In contexts where innovative ideas, initiatives and transactions proliferate (Chenhall and Morris 1995), aligning DDCs with digital transformation strategy becomes crucial.

The interactive use of MCSs operates by fostering continuous dialogue, learning, and adaptation, supporting the exploration of new opportunities, dynamic strategy adjustments, and effective management of uncertainties, such as changes in competition, technological advancements, or changing customer preferences. Rather than adhering strictly to predefined targets, interactive use encourages experimentation and innovation. It also acts as an attention-shaping device, directing discussion toward strategically relevant issues while filtering out less relevant signals, thereby concentrating effort on strategically aligned goals (Bisbe and Otley 2004; Bisbe and Malagueño, 2015). Such a filtering role becomes particularly relevant in digital transformation contexts, where managers face an abundance of signals of varying importance. As previously mentioned, DDCs involve scanning for trends, seizing opportunities and redesigning internal structures. The interactive use of MCSs enables organizations to respond swiftly to digital opportunities by providing real-time monitoring and feedback. This agility allows firms to adjust strategies more effectively, thereby enhancing their competitive position. Interactive MCSs also facilitate trend scanning, structural redesign, and quicker compliance with customer and partner demands. This, in turn, contributes to an improved corporate image and increased organizational value.

Accordingly, the following hypothesis is formulated:

H.4. *The emphasis that firms place on the interactive use of MCSs positively moderates the relationship between DDCs and DTS.*

The purpose of a belief system is to encourage subordinates to adopt values and objectives espoused by top management (Widener 2007). When managers cultivate a digital mindset, belief systems play a crucial role in fostering commitment to digital initiatives by communicating what is important for the company. Belief systems create a shared sense of purpose, encouraging employees to align their actions with corporate objectives and facilitating the integration of DDCs into organizational strategies through the internalization of their values and goals (Bedford 2015). This is particularly relevant in the context of DDCs, as they involve sensing opportunities, seizing them according to corporate mission, vision and values and implementing them through long-term strategic actions. By embedding digital initiatives into daily operations, belief systems reduce resistance to new digital tools, resulting in improved efficiency and ultimately strengthening a competitive advantage, a positive corporate image and the value of the organization. From a managerial perspective, a strong alignment between digital initiatives and organizational values enhances DTS.

Consequently, the following hypothesis is formulated:

H.5. *The emphasis that firms place on belief systems positively moderates the relationship between DDCs and DTS.*

4 Methodology and data

4.1 Data collection

The sample consisted of 415 managers from Spanish and Colombian organizations with at least 10 employees and a three-year period of operation. Both countries share Spanish as the most widely spoken language with some slight differences; therefore, the vast majority of the respondents were developing their functions in Spanish or Colombian companies. For this reason, they were contacted in the Spanish language. The data collection was conducted between May and July 2021. An online questionnaire was used for information gathering. Although the procedures used in Spain and Colombia were quite similar, they were not identical. We targeted companies in Spain and Colombia with the support of institutions that promote digital transformation. Owing to the low response rate, we broadened our strategy to include more companies, always taking measures at every step to ensure that they were engaged in a digital transformation process, with the appropriate explanation in the questionnaire used for data collection. Thus, the final sample was composed only of organizations following a digital transformation process.

In Spain, an agreement was signed with the Alastria Network Consortium, which comprises 554 entities involved in fostering the use of blockchain, a technology that has recently been associated with digital transformation projects. Alastria initially disseminated the questionnaire by including it in two consecutive newsletters. Owing to the insufficient number of responses obtained (our target was 200 responses, but we obtained only 9), we extracted the list of Spanish companies with more than 10 employees from the SABI database.[2] We sent many messages to these companies, which considerably increased the response rate. We finally disseminated the questionnaire following a LinkedIn search for people whose job title was CIO, CTO or CDO and whose current organization was an established company based in Spain with more than 10 employees.

Contact with Colombian companies was first made through the Colombian research team, who contacted the *Centro de Transformación Digital Empresarial (CTDE*), which comprises 3,300 entities that are strongly concerned with digital transformation. The response rate was extremely low. We subsequently contacted Colombian companies after conducting a LinkedIn search on the basis of the same criteria as those used in the Spanish search. The results were much poorer than those in Spain. Therefore, given the very low response rate in Colombia, three Colombian researchers were hired to conduct fieldwork in the country. This approach yielded

[2] SABI (Iberian Balance Sheet Analysis System) is a database compiled by INFORMA D&B in collaboration with Bureau Van Dijk that includes general information regarding and annual accounts of Spanish and Portuguese companies.

very good results, as the response rate increased considerably, becoming equal to the rate achieved in Spain.

All these efforts resulted in a total of 665 participants who initiated responses to the questionnaire, from whom we were ultimately able to obtain 415 valid responses to the full questionnaire, 209 from Spain and 206 from Colombia, yielding response rates of 11.04% and 10.33%, respectively. The characteristics of the respondents and their companies are shown in Table 1.

To ensure that our sample of organizations did not suffer from unit response bias, we implemented random sampling in our database. Moreover, to assess potential bias across countries, an ANOVA was conducted. The results indicated that there were no significant differences ($p > 0.05$) in the mean values of any item in the questionnaire, regardless of the country from which the responses were obtained. Therefore, it can be concluded that the country of origin did not introduce any bias into the sample.

A firm is identified as a family business when participation by the ownership family represents at least 50% of the firm's capital.

Table 1 Sample characteristics

Respondent characteristics	
Gender	
Male	65.06%
Female	33.73%
Position	
Chief executive officer	10.60%
Controller	10.12%
Chief information officer (CIO)	4.58%
Chief technology officer (CTO)	2.41%
Chief digital officer (CDO)	10.84%
Chief operations officer (COO)	1.20%
Chief financial officer (CFO)	13.01%
Chief marketing officer (CMO)	16.63%
Chief communications officer (CCO)	10.60%
Other	20.00%
Average age and tenure	
Average age	37.32
Average tenure at the company	6.54
Average tenure at the present position	4.21
Company characteristics	
Public vs. private	
Private company	94.85%
Public company	5.15%
Family vs. nonfamily	
Family firm	31.58%
Nonfamily firm	67.67%
Individual vs. group	
Individual company	55.72%
Member of a company group	44.53%
Location	
Spain	50.36%
Colombia	49.64%

4.2 Variable measurement

4.2.1 Management control systems

Several items were used to measure the four levers of control, which account for the interactive use of MCSs and belief systems as the main variables of the model and the diagnostic use of MCSs and the boundary system as control variables. The item formulation was inspired by previous studies (Bisbe and Malagueño 2015; Gomez-Conde et al. 2019; Henri 2006a; Naranjo-Gil 2016; Widener 2007). The respondents were asked to indicate the extent to which they agreed with these items on a five-point Likert scale ranging from 1 (not at all) to 5 (a great deal) (see the Appendix).

4.2.2 Digital dynamic capabilities

To assess organizations' digital capabilities, this study used the framework developed by Warner and Wäger (2019). On a 5-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree), the respondents indicated their level of agreement with statements aimed at assessing the existence of these capabilities in their companies. This framework was inspired by Teece (2007), and it was also used by Vial (2019), Li and Liu (2014) and North et al. (2019) (see the Appendix).

4.2.3 Digital transformation success

On a 5-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree), the respondents indicated their level of agreement with statements referring to the achievement of company objectives resulting from digital transformation implementation. Four elements were considered: 1. competitive position; 2. the organization's image among different stakeholders (suppliers, clients, shareholders, workers, etc.); 3. organizational value growth; and 4. degree of achievement in relation to expectations. This type of self-reported subjective assessment of performance is widely used in research (White et al. 2022), and it is particularly appropriate when the observations are from different countries and sectors (Singh et al. 2016), as in this case (see the Appendix).

4.2.4 Control variables

The diagnostic use of MCSs and boundary systems were included in our study as control variables to improve the validity of our results by considering Simons' four levers of control in our model. Other control variables were also incorporated. According to Berbel-Vera et al. (2022), the role of the CDO is strongly related to the development of DDCs and the success of the digital transformation process. Nevertheless, not every company includes this specific position, and other profiles in the organization may be responsible for the functions attributed to a CDO. Accordingly, respondent position was included as a control variable (Holm and Ax 2020), and respondents were grouped on the basis of the following criteria. Group 1: profiles directly engaged in digital transformation. This group was characterized by transver-

sal competences associated mainly with IT (CEO, controller, CIO, CTO or CDO), all of which exhibited strong engagement with digital transformation processes. Group 2: functional leaders (users of technology but not directly engaged in digital transformation of the organization). Group 2 is more closely associated with specific competences related to particular areas of knowledge; we refer to chief operations officers (COOs), chief financial officers (CFOs), chief marketing officers (CMOs), chief communications officers (CCOs) and other positions that are equivalent but not focused on IT beyond a user role. This categorization[3] is consistent with the suggestions of Kaplan and Sorensen (2021). In addition, according to research on MCSs, the following control variables were included: organization size and industry (operationalized as two dummy variables: Dummy 1 Secondary sector and 0 Primary and Tertiary sectors and Dummy 1 Tertiary sector and 0 Primary and Secondary sectors) (Bedford et al. 2019; Holm and Ax 2020); country (Colombia: 1; Spain: 2) (Holm and Ax 2020); private vs. public (private: 1; public: 2) (Henri and Wouters 2020; Li et al. 2020); family vs. nonfamily (family: 1; nonfamily: 2) (Guenther and Heinicke 2019); and business group vs. individual (business group 1; individual:[4] 2) (Bisbe and Otley 2004).

5 Findings

Given that our data for both the predictor and criterion variables were gathered from single respondents using consistent scale formats, we examined the potential impact of common method bias. To do this, we employed Harman's single-factor test, a widely used diagnostic procedure for this purpose (Podsakoff et al. 2003). The test involves running an unrotated exploratory factor analysis on all relevant measurement items simultaneously to determine if a single factor accounts for the majority (over 50%) of the total variance.

We performed this test twice. First, for the 18 items measuring the DDCs construct (detailed in Table 2), the initial unrotated factor captured 46.74% of the variance. Second, when applied to all the items used in the validation of our main study constructs (Table 4), the first factor accounted for 38% of the variance.

Since neither of these percentages exceeds the common 50% threshold, the results suggest that a single, general factor does not explain the bulk of the covariance among the measures. While Harman's test is not definitive, this provides some confidence that common method bias is unlikely to be the primary driver of the findings reported in this study.

[3] The positions labelled "directly engaged in digital transformation" and "functional leaders" were reconsidered for the purposes of this research, but other labels could also be possible

[4] A single company is understood to be an independent firm, while a business group is understood to be a corporation composed of several single companies.

Table 2 Analysis of the dimensionality, reliability, and validity of the scale of DDCs (fully standardized solution)

Item	Factor loading	(t value)
Digital sensing (CR = 0.94; AVE = 0.77; Cronbach's α = 0.941)		
Screening of digital competitors	0.81	(10.93)
Sensing customer-centric trends	0.81	(11.14)
Analysing scouted signals	0.89	(11.66)
Interpreting future digital scenarios	0.87	(10.80)
Establishing a long-term digital vision	0.88	(12.50)
Promoting a digital mindset	0.87	(10.61)
Digital seizing (CR = 0.94; AVE = 0.76; Cronbach's α = 0.940)		
Creating minimum viable products	0.82	(11.20)
Considering a lean start-up methodology	0.85	(12.02)
Scaling up innovative business models	0.86	(13.77)
Setting an appropriate speed of execution	0.88	(11.75)
Pacing strategic responses	0.83	(7.92)
Accepting redirection and change	0.87	(9.38)
Digital transforming (CR = 0.93; AVE = 0.73; Cronbach's α = 0.928)		
Interacting with multiple external partners	0.78	(10.09
Exploiting new ecosystem capabilities	0.88	(10.20)
Hiring a chief digital officer	0.80	(10.16)
Digitalization of business models	0.86	(13.43)
Identifying digital workforce maturity	0.84	(9.20)
External recruitment of digital talent	0.81	(10.81)

Fit of the model: Chi-squared = 154.8937, df = 129, p = 0.05983; RMSEA = 0.022; CFI = 0.995; and NNFI = 0.994

CR = Composite reliability

AVE = Average variance extracted

5.1 Validation of the DDCs scale and item parcelling

As a first step, to measure DDCs, we employed a scale that was based on Warner and Wäger's (2019) previous qualitative work and was developed for this project.[5] Thus, we worked with a validated construct consisting of three dimensions, i.e., digital sensing, digital seizing and digital transforming, with a total of 18 items (6 items per dimension).

Descriptive analysis and confirmatory factor analysis were conducted to assess the reliability and validity of the entire DDCs scale, as Tables 2 and 3 illustrate. As shown in Table 2, the composite reliabilities (CRs) (ranging from 0.93 to 0.94) and all the average variance extracted (AVE) values (ranging from 0.73 to 0.77) were greater than the 0.5 cut-off. In addition, all the Cronbach's alpha coefficients per dimension (ranging from 0.928 to 0.941) exceeded the accepted reliability threshold of 0.70. Thus, all the measures demonstrated adequate reliability.

Table 2 also shows that confirmatory factor analysis (CFA) indicates that the model fits the data well since CFI = 0.995 and NNFI = 0.994, both of which exceed 0.99; in addition, the RMSEA did not exceed 0.08. All item loadings, which ranged from 0.77 to 0.89, were significant at the 0.001 level, indicating that the convergent validity of all the measures was acceptable.

[5] The previously conducted analyses used to develop this scale are undergoing revision for forthcoming publication.

Table 3 DDC constructs: descriptive analysis, correlations, reliabilities, and discriminate validities of the measures

Model constructs	Digital sensing	Digital seizing	Digital transforming
Digital sensing	**0.88**		
Digital seizing	0.69	**0.87**	
Digital transforming	0.69	0.53	**0.86**
Mean	3.84	3.62	3.59
SD	0.96	1.01	1.07

Diagonal elements (in bold) represent the square root of AVE. Below the diagonal: estimated correlations among the factors

SD: Standard deviation.

According to Jöreskog and Sörbom (1996), an overall good fit of the scale is indicated when the probability associated with the chi-square test reaches a value higher than 0.05 (0.05983), as shown in Table 2.

According to Table 3, all diagonal elements representing the square roots of the AVEs are larger than any other corresponding row or column entry, indicating that each construct differs from the other constructs sufficiently; therefore, the discriminant validities of all the measures are established.

As a second step, before validating all the scales, we performed item parcelling by calculating the average of the items sharing the same dimension in the DDCs construct to form composite measures (Bandalos and Finney 2001; Bou-Llusar et al. 2009; Landis et al. 2000). Composite measures of DDCs are combinations of items used to create an aggregated score that is then subjected to CFA alongside the remaining scales considered in the study with the aim of validating them (see Table 4). In CFA, composite measures are useful for two reasons. First, they enable us to meet the normal-distribution assumption of maximum likelihood estimation more effectively. Second, they result in more parsimonious models because this approach reduces the number of variances and covariances that must be estimated, thus increasing the stability of the parameter estimates, improving the variable-to-sample-size ratio and reducing the impact of sampling error on the estimation process (Bagozzi and Edwards 1998; Bandalos and Finney 2001; Little et al. 2002). Additionally, for the DDCs scale, the invariance of the instrument of measurement was verified, thereby ensuring comparability among the estimated parameters that used composite measures as inputs. Thus, a composite measure for each DDC dimension was introduced as an indicator variable for the analyses conducted to assess the dimensionality, reliability and validity of the scales. The invariance of the measurement instrument was subsequently verified to compare the regression coefficients of the model.

Exploratory factor analyses (EFA) were conducted to verify the factor structures in both steps of the validation process. The first EFA, performed on the initial DDCs scale, yielded a Kaiser–Meyer–Olkin (KMO) measure of 0.970 and a significant Bartlett's test of sphericity ($p<0.001$). This analysis explained 77% of the total variance, with all factor loadings exceeding 0.7. Similarly, the second EFA, which examined the main constructs of the model (including the DDCs scale with parcelled items), resulted in a KMO of 0.952 and a significant Bartlett test ($p<0.001$). It accounted for 79.1% of the explained variance, and again, all factor loadings were above 0.7. In both instances, the EFA results supported the factor structures identified.

Table 4 Analysis of the dimensionality, reliability, and validity of the scale (fully standardized solution)

Item	Factor loading	(t value)
iMCS (CR = 0.93; AVE = 0.72; Cronbach's α = 0.925)		
To enable discussion in meetings among superiors, subordinates and colleagues	0.773	(10.74)
To promote the continuous discussion of results, relevant data and action plans	0.828	(10.00)
To provide a common vision for the organization	0.859	(11.48)
To develop a common vocabulary within the organization	0.804	(10.22)
To enable you to focus on issues that affect the entire organization	0.863	(10.56)
To coordinate the different units in the organization with one another	0.800	(8.31)
Belief (CR = 0.90; AVE = 0.74; Cronbach's α = 0.904)		
The organization's mission statement clearly communicates the company's core values to employees	0.845	(6.07)
Senior management communicates core values to employees	0.840	(10.06)
Employees are aware of the organization's core values	0.822	(9.25)
The organization's mission inspires employees	0.846	(9.13)
DDCs (CR = 0.93; AVE = 0.83; Cronbach's α = 0.928)		
Digital sensing	0.888	(6.57)
Digital seizing	0.910	(6.00)
Digital transforming	0.905	(7.86)
DTS (CR = 0.92; AVE = 0.77; Cronbach's α = 0.916)		
It is improving the competitive position in the environment in which your organization operates	0.888	(10.45)
It is improving the image of the organization among different stakeholders (suppliers, clients, shareholders, workers, etc.)	0.848	(12.20)
It is enhancing the value of the organization for its members, partners, shareholders and/or investors	0.825	(7.79)
The expectations of your organization regarding the digital transformation are being met	0.862	(11.26)

Fit of the model: Chi-squared = 114.5783, df = 109, p = 0.33851; RMSEA = 0.011; CFI = 0.998; and NNFI = 0.998

CR = Composite reliability. AVE = Average variance extracted. iMCS: Interactive use of MCSs; Belief: belief systems; DTS: Digital transformation success.

5.2 Validation of all scales

After composite measures of items sharing the same dimension in the DDCs scale were formed, following Bandalos and Finney (2001), Bou-Llusar et al. (2009) and Landis et al. (2000), we analysed the psychometrical properties of all scales included in the model. Convergent validity was demonstrated because the factor loadings were greater than 0.5 (Bagozzi 1980; Bagozzi and Yi 1988; Hair et al. 2006); in addition,

convergent validity was evident because the AVEs (ranging from 0.72 to 0.83) for each of the factors were greater than 0.5 (Fornell and Larcker 1981). With respect to the reliability of the scale, the indices of the CRs (ranging from 0.90 to 0.93) of each of the dimensions obtained were greater than 0.7 (Bagozzi and Yi 1988). Table 4 also shows that the Cronbach's alpha coefficients for all the constructs detailed (ranging from 0.904 to 0.928) exceeded the accepted reliability threshold of 0.70. Thus, all the measures demonstrated adequate reliability.

Table 4 also illustrates that CFA indicates a model that fits the data well since CFI = 0.998 and NNFI = 0.998, both of which are close to 1; in addition, the RMSEA does not exceed 0.08. All item loadings, ranging from 0.773 to 0.910, were significant at the 0.001 level, indicating that the convergent validities of all the measures were acceptable.

Finally, in Table 5, the discriminant validity of all the constructs can be checked by considering the AVE (Fornell and Larcker 1981) when each construct shares more variance with its indicators than with other constructs in the model. Discriminant validity can be verified when the square root of the AVE between each pair of constructs is greater than the estimated correlation between those constructs, which is the case in our model.

5.3 Results of the test of the research hypotheses

The proposed models were analysed via regression equations, which generally allow for the testing of the relationships between dependent and independent variables simultaneously. EQS was used to apply the robust maximum likelihood (MLR) method, as the missing values for some variables had to be accounted for.

The hypotheses proposed in the empirical section were then tested. The equations that enabled us to contrast the hypotheses are shown below:

$$\begin{aligned} DDC_i = & \beta_0^1 + \beta_1^{H1} iMCS_i \\ & + \beta_2^{H2} Belief_i + \gamma \sum controls + \varepsilon_i^1 \end{aligned} \tag{1}$$

$$DTS_i = \beta_0^2 + \beta_1^{H3} DDC_i + \gamma \sum controls + \varepsilon_i^2 \tag{2}$$

Table 5 Descriptive analysis, correlations, reliabilities, and discriminant validities of the measures for all the scales

Model constructs	iMCS	Belief	DDCs	DTS
iMCS	**0.85**			
Belief	0.66	**0.86**		
DDCs	0.51	0.61	**0.91**	
DTS	0.49	0.62	0.67	**0.88**
Mean	3.89	4.03	3.68	3.91
SD	0.91	0.88	0.95	0.92

Diagonal elements (in bold) represent the square root of AVE. Below the diagonal: correlations estimated between factors

$$\begin{aligned} DTS_i = & \beta_0^3 + \beta_1^{H3} DDC_i \\ & + \beta_2 iMCS_i \\ & + \beta_3^{H4} DDC_i \ x \ iMCS_i \\ & + \beta_4 Belief_i \\ & + \beta_5^{H5} DDC_i \ x \ Belief_i \\ & + \gamma \sum controls + \ \varepsilon_i^3 \end{aligned} \tag{3}$$

Acknowledging the prevalence of deviations from the multivariate normality assumption within social science data—a prerequisite for precise inference via standard maximum likelihood (ML) estimation—the estimation of the models presented in Eq (1), (2), and (3) was undertaken utilizing the robust maximum likelihood (MLR) methodology available within the EQS 6 programme (Bentler 2006). It is well established that violation of the normality assumption can yield inflated chi-square statistics and underestimated standard errors when the standard ML approach is employed (e.g., Curran et al. 1996; Hu et al. 1992). To address this limitation, the MLR approach within EQS incorporates the Satorra-Bentler scaling correction (Satorra and Bentler 1988, 1994) to the chi-square statistic, alongside the computation of robust standard errors derived from the 'sandwich' estimator (Bentler and Dijkstra 1985; Browne 1984). This combined approach has been empirically demonstrated to furnish more precise and reliable statistical inferences under various conditions of nonnormality (e.g., Chou et al. 1991; Fouladi 2000), thereby enhancing the validity of both parameter significance tests and the overall model goodness-of-fit assessment.

Table 6 displays the regression results of our models, accounting for both the main and the moderating effects.

Regarding the main effects, the results reveal significant relationships across the corresponding regressions. H1 and H2 are strongly supported; both the interactive use of MCSs ($\beta_1^{H1} = 0.355$***, $p<0.001$) and belief systems ($\beta_2^{H2} = 0.279$***, $p<0.001$) exhibit positive associations with DDCs. Although both the diagnostic use of MCSs and boundary were included in the analysis of main effects, only the diagnostic use of MCSs showed a significant positive association with DDCs ($\beta=0.167$**, $p<0.01$). This finding suggests a potentially relevant aspect of how MCS levers operate during digital transformation processes. H3 is also validated, as DDCs have a robust positive effect on DTS ($\beta_1^{H3} = 0.789$***, $p<0.001$). Additionally, boundary systems exhibit, albeit weaker, a positive relationship with DTS ($\beta=0.087$*, $p<0.05$).

Turning to the moderating effects, the results indicate that H4 is not supported, as the interaction term DDCs × iMCS ($\beta_3^{H4} = 0.071$, $p>0.05$) has no significant influence on DTS. This implies that although the interactive use of MCSs independently enhances DDCs (as shown in H1), it does not strengthen the relationship between DDCs and DTS. In contrast, H5 yields unexpected results: On the one hand, the interaction DDCs × Belief ($\beta_5^{H5} = -0.393$*, $p<0.05$) is significant but negative, contradicting the hypothesized positive moderation proposed. This suggests that as DDCs increase, the influence of belief systems on DTS diminishes. On the other

Table 6 Structural equation modelling results: main and moderating effects

	Main effects	Moderating effects	
Hypotheses	(1) H1 & H2	(2) H3	(3) H4 & H5
Dependent variables	DDCs	DTS	DTS
Independent variables			
Constant	3.682***	3.931***	3.938***
Main effects			
iMCS	0.355***		
Belief	0.279***		
DDCs		0.789***	0.890***
Moderating effects			
iMCS			−0.061
DDCs x iMCS			0.071
Belief			0.236*
DDCs x belief			−0.393*
Control variables			
Respondent position	0.080*	−0.064*	−0.056*
lnSize	0.016	−0.017	−0.031
Country (1: Colombia; 2: Spain)	0.119**	−0.026	−0.026
Dummy 1 secondary sector and 0 primary and Tertiary sectors	0.106	0.044	0.055
Dummy 1 tertiary sector and 0 primary and Secondary sectors	0.210	0.038	0.045
Private (1) vs. public (2)	0.011	0.020	0.027
Family (1) vs. nonfamily (2)	0.002	0.015	0.013
Business group (1) vs. individual (2)	−0.022	0.035	0.034
dMCS	0.167**	0.042	0.035
Boundary	−0.015	0.087*	0.105**
R^2	0.547	0.715	0.721
χ^2	12.7306	10.2309	22.8405
df	11	10	14
p	0.31130	0.42048	0.06292
NFI	0.986	0.984	0.994
CFI	0.985	0.983	0.994
Standardized RMR	0.052	0.039	0.030

dMCS: diagnostic use of MCSs; Boundary: boundary systems.

*$p<0.05$; **$p<0.01$; ***$p<0.001$; ns = not significant.

hand, boundary systems appear to play a significant role in relation to DTS, as evidenced by the results for H4 and H5 (β = 0.105**, $p<0.01$).

A set of additional control variables was included in the analysis. The respondent position shows a positive and significant relationship for H1 and H2 (β = 0.080*, $p<0.05$), indicating that higher levels of DDCs are perceived in more specialized positions (Group 2: Functional leaders). In contrast, a significant but negative relationship was found for H3, H4 and H5, revealing that, the more transversal the position (Group 1: Users directly engaged in digital transformation), the stronger the perception of DTS (β = −0.064* to −0.056*, $p<0.05$), compared with those in spe-

Table 7 Comparison of the standardized coefficients (β) between the Colombian and Spanish subsamples (effect of iMCs/Belief on DDCs)

Independent variable	Colombian subsample (β, SE)	Spanish subsample (β, SE)
iMCS	0.371 (0.46)***	0.742 (0.048)***
Belief	0.327 (0.41)***	0.152 (0.060)*

Dependent variable: DDC.

SE = standard error.

*$p<0.05$; **$p<0.01$; ***$p<0.001$; ns = not significant.

Table 8 Comparison of the standardized coefficients (β) between the Colombian and Spanish subsamples (effect of DDCs on DTS)

Independent variable	Colombian subsample (β, SE)	Spanish subsample (β, SE)
DDCs	0.846 (0.027)***	0.795 (0.024)***

Dependent variable: DTS.

SE = standard error.

*$p<0.05$; **$p<0.01$; ***$p<0.001$; ns = not significant.

cialized positions (Group 2). Finally, the country variable (β = 0.119**, $p<0.01$) has a significant positive effect on DDCs. This suggests that organizations in Spain (coded as 2) exhibit stronger DDCs than those in Colombia (coded as 1).

All the models demonstrate robust fit indices. The χ^2 test yields p values >0.05, indicating nonsignificant deviations from the expected covariance matrices. The NFI and CFI values (0.983–0.994) closely approximate unity, exceeding the recommended threshold of 0.90, whereas the standardized RMR (0.030–0.052) remains near zero, confirming minimal discrepancies between the observed and predicted data (Jöreskog and Sörbom 1996). These metrics collectively validate the structural models' reliability and adequacy for representing the hypothesized relationships.

5.4 Robustness checks

In addition to the above empirical analyses using regression equations and EQS, we also conducted various robustness checks to minimize measurement errors and control for potential confounding effects. Specifically, we examined whether the observed relationships remained consistent across subsamples by splitting the data between Colombian and Spanish respondents and comparing standardized coefficients.

First, we analysed the direct effects of iMCs and Belief on DDCs. As shown in Table 7, the relationships remained significant in both subsamples, providing additional support for the stability of the findings.

Second, we verified the direct effects of DDCs on DTS. Again, the significance of the relationships was maintained across both subsamples (see Table 8), reinforcing the robustness of our results.

6 Discussion

The study confirms the expected main effects of the hypothesized MCSs and, additionally, identifies a particularly significant role of the diagnostic use of MCSs in the development of DDCs. The results strongly support H1 and H2 and align with recent levers of control studies that emphasize the importance of enhancing organizational capabilities to improve performance (Bellora-Bienengräeber et al. 2023), even though we considered digital nontraditional capabilities in this study.

These findings are consistent with previous studies on the interactive use of MCSs and innovation (Bisbe and Otley 2004; Bisbe and Malagueño 2015; Moulang 2015; Lopez-Valeiras et al. 2016) and align with theoretical expectations. The interactive use of MCSs facilitates cross-functional collaboration, knowledge sharing, and adaptive resource allocation (Bisbe and Otley 2004; Müller-Stewens et al. 2020), whereas belief systems foster a shared digital mindset, promote experimentation and enhance digital strategic alignment (Autio et al. 2018; Janka et al. 2020), all of which contribute to the enhancement of DDCs in the context of digital transformation.

However, some additional results were also obtained since the diagnostic use of MCSs was found to be significantly associated with DDCs. Our findings suggest that monitoring outcomes and correcting deviations from expectations—i.e., the diagnostic use of MCSs—support the promotion of DDCs. While some studies have reported that the diagnostic use of MCSs can constrain innovation (Henri 2006a; Bisbe and Malagueño 2015), others have reported their positive contribution (Koufteros et al. 2014; Bedford 2015). Assuming that in the context of digital transformation, the diagnostic use of control systems might be business intelligence-enabled, our findings align with those of Elbashir et al. (2021), who confirmed the positive influence of the interaction between information systems and business intelligence-enabled MCSs on organizational performance. This may be due to the evolution of diagnostic MCSs, which now go beyond control functions to support forecasting, opportunity detection, and strategic decision-making in digital environments (Strand and Syberfeldt 2019). Moreover, our findings also align with recent studies on the impact of MCSs on innovation and creativity. For example, Chen et al. (2023) demonstrated that the combined use of diagnostic and interactive controls can enhance product innovation, whereas Appuhami et al. (2024) reported that the diagnostic use of budgets positively contributes to managers' creativity.

This result underscores the importance of integrating diagnostic MCSs into the digital transformation discourse, as they support the monitoring and alignment mechanisms essential for implementing DDCs effectively. This finding challenges the assumption that only interactive MCSs drive digital agility, suggesting instead that a balanced deployment of both diagnostic and interactive controls may be most effective in supporting DDCs development.

For the moderating effects, two variables were tested, but only one was confirmed: the interactive use of MCSs did not have a significant moderating effect, and the moderating role of belief systems was unexpectedly negative. In contrast, boundary systems emerged as a significant driver. Although the interactive use of MCSs directly contributes to the development of DDCs, their lack of a significant moderat-

ing effect[6] may reflect the influence of other contextual factors that may limit their ability to strengthen other relationships. This highlights the need for an alignment between control practices and the organization's digital strategy to fully leverage their potential in complex environments.

In this context, the analyses of the moderating effects revealed the paramount role of value systems (belief and boundary), rather than the type of MCS used (interactive and diagnostic). Specifically, belief systems were found to negatively moderate the relationship between DDCs and DTS. Although belief systems emerged as a significant predictor of DDCs in the regression analysis, the negative coefficient observed in the moderation of DDCs and DTS suggests that, in certain contexts, a greater emphasis on belief systems may be associated with lower levels of DTS. This paradox can be explained by the stage of digital capability development. In the early phases of DDCs, belief systems reinforce shared values that foster alignment, reduce uncertainty, and enable coordinated action (Bartlett and Ghoshal 1993; Jansen et al. 2008; Henri 2006b; Simons 1995). However, as DDCs mature and become embedded in the organization's strategy, the overly intensive use of belief systems may become less effective, potentially leading to employee fatigue or limiting flexibility. In such cases, overemphasis on shared assumptions may hinder the adaptability required for advanced digital transformation (Warner and Wäger 2019; Janka et al. 2020). The negative coefficient, therefore, does not reflect an inherently harmful role of belief systems but rather highlights the importance of dimensioning, timing and context.

Regarding boundary systems, a significant role has been detected both for direct effects (Eq 2) and for moderating effects (Eq 3), where they were included as a control variable. This result suggests that digital transformation is more successful when DDCs operate within clearly defined limits. Boundary systems—formal rules and ethical guidelines—enhance DTS by reducing risks, clarifying strategic priorities, and preventing resource misallocation. By setting clear operational boundaries, they reduce ambiguity and ensure alignment with organizational objectives such as competitive advantage, stakeholder value, and expectation fulfilment (Simons 1995; Mundy 2010). These findings support the view that boundary systems complement agile digital strategies by balancing innovation with disciplined execution (Marginson 2002; Bisbe and Malagueño 2015). Although prior research focused on value systems in innovation processes (Bisbe and Malagueño 2015), our study highlights the distinct relevance of boundary systems in the context of digital transformation.

In addition, the respondent´s position within the organization plays a key role in the perception of the digital transformation process, aligning with prior findings on CDO's role (Berbel-Vera et al. 2022). Specifically, respondents from Group 2 (functional leaders) reported higher perceptions of DDCs. This may reflect an over-

[6] In the previous analysis, we assessed the moderating effects of iMCS and belief systems separately through distinct regression models to isolate each interaction. In a second step, we re-estimated the model via structural equation modelling (SEM) with robust maximum likelihood estimation (MLR) in EQS, incorporating both moderators and their interactions into a single equation. This approach provides a more robust estimation by accounting for correlations among predictors and addressing nonnormality. In the revised model, only the interaction between DDCs and belief systems remained significant, suggesting that belief systems play a more prominent moderating role when both contextual variables are considered jointly.

estimation driven by their functional expertise, which emphasizes domain-specific digital tools (e.g., marketing analytics, financial systems) while overlooking cross-functional integration and broader strategic alignment (Berbel-Vera et al. 2022). However, the respondents in Group 1 (directly engaged in the digital transformation process) perceived a greater effect of DDCs on DTS than did those in Group 2. This difference may stem from their broader perspective, as DTS goes beyond financial results to include dimensions such as competitive positioning, market image, stakeholder value, and ecosystem integration (Kaplan and Sorensen 2021).

We also acknowledge that the country context can influence the technological growth potential. The positive and significant effects of this factor on DDCs reflect varying degrees of technological development across countries. The gap observed between Spain and Colombia may be explained by Spain's institutional and infrastructural advantages, such as mature digital ecosystems, supportive regulation, and greater access to technological resources, which strengthen firms' capacity to sense, seize, and transform digital opportunities (Bisbe and Otley 2004; Teece 2007). Nevertheless, the country does not appear to influence the moderating regressions.

Finally, to ensure the robustness of our findings, we conducted country-specific analyses for Spain and Colombia. Results were largely consistent with the overall sample, though some differences emerged: iMCS showed a stronger effect on DDCs in Spain, while belief systems were more significant in Colombia. These variations may be linked to cultural dimensions (Hofstede's framework, 1980): Spain's individualism and high uncertainty avoidance strengthen the role of interactive controls, whereas Colombia's collectivism and higher power distance reinforce the impact of shared values. While cross-country comparisons are not the primary focus of this study, the findings indicate contextual differences but confirm that the overall conclusions hold across countries.

7 Conclusions

The digital transformation process presents both significant challenges and valuable opportunities for organizations. MCSs can play a critical role in facilitating the success of this transformation process. Drawing on Warner and Wäger's (2019) DDCs framework and Simons' (1995) levers of control, this study aimed to achieve two objectives. First, it sought to identify the type of MCS that can enhance the development of DDCs, a key antecedent of successful digital transformation. Second, it aimed to investigate whether any MCS levers of control also moderate the impact of DDCs on DTS.

In relation to the first objective, we concluded that while the most significant effects are attributed to the interactive use of MCSs and belief systems, both of which enhance flexibility, adaptability and confidence, managers should also recognize the potential value of diagnostic systems in promoting DDCs.

In addition, to enhance the relationship between DDCs and DTS, managers should reinforce the implementation of value systems (belief and boundary) rather than focus on how to use MCSs. Therefore, in this context, attention should be given to the intensity of use of belief systems in combination with boundary systems, whose role

in the digital transformation process was revealed to have a significant and positive impact on DTS. As previously argued, more intensive use of belief systems does not necessarily imply that DDCs have a stronger effect on DTS. Contrary to expectations, the negative moderation found regarding belief systems encouraged us to consider boundary systems as potentially supportive of DTS.

These results also have valuable managerial implications, as they demonstrate that the style of MCS use actually matters for reaching DTS. For companies involved in this process, two key considerations emerge from the tested relationships. First, the findings suggest that implementing each lever of control in isolation has limited relevance in the context of digital transformation. Instead, identifying the optimal balance that best promotes the development of DDCs should be a priority for digital transformation leaders. This can be achieved by encouraging open face-to-face dialogue and debate among superiors, subordinates and peers and by fostering a supportive environment that facilitates information sharing, i.e., an environment that avoids rigid structures and the strict following of rules. Moreover, managers should not overlook the benefits of diagnostic controls, which play a significant role in fostering DDCs through their monitoring function. It is also essential that companies promote values aligned with DDCs development; otherwise, difficulties will arise in their attempts to digitally transform the company. Second, managers must acknowledge that a well-established belief system, once a certain level of DDCs is achieved in the organization, is not enough to consolidate a successful digital transformation. Organizations must balance the intensity of the use of belief systems with the use of boundary systems, which helps focus on the new objectives, thereby minimizing resource waste and avoiding strategy misalignment. By effectively managing these control elements, organizations can strengthen their digital transformation efforts and enhance the overall success of the process.

Our first theoretical contribution is the examination of the role of MCSs in eliciting DDCs, which extends Simons' levers of control framework by introducing the uses of MCSs that can have different impacts on the digital transformation process. Our second contribution lies in identifying the managerial control practices that support the digital capabilities crucial for these processes, thus extending the emerging DDC literature. Finally, our third contribution consists of empirically demonstrating the relationship between DDCs and DTS in organizations empirically, which was accomplished by referencing the results of the statistical tests conducted.

This study has two main limitations that offer ample opportunities for future research. First, DTS was measured using self-report survey responses, which were based on subjective perceptions rather than objective financial indicators, a common approach in prior research on dynamic capabilities. Second, the data were collected from a single respondent per organization, which may not fully capture the diversity of perspectives within each firm. Future studies should consider multi-informant designs and objective performance data to strengthen validity.

Future research could explore the role of diagnostic MCSs and boundary systems in greater depth, particularly in relation to their relevance in digital transformation processes. A deeper examination of how these control systems interact with the sensing, seizing, and transforming subcapabilities of dynamic capabilities would also provide valuable insights. To deepen the understanding of digital dynamic capabili-

ties, future research could analyse the sensing, seizing, and transforming subcapabilities individually to uncover differences based, for example, on key variables such as sector, geographic location or organization size. Cross-national studies comparing countries with different cultural contexts may reveal how cultural values influence the design and use of MCSs and the effectiveness of digital transformation strategies. Similarly, research focusing on specific sectors with more homogeneous characteristics could uncover sector-specific patterns and challenges. Additionally, the relevance of the respondents' profiles suggests that the evolution of leadership roles, including digital skills, strategic vision, and data literacy, warrants closer examination. Understanding how leaders shape and drive digital transformation efforts will be critical to advancing both theory and practice in this field.

Appendix

Survey questions

Interactive (iMCS) and diagnostic (dMCS) use of management control systems
To what extent does the top management team use management control systems? (1: Not at all, 5: A lot)
… To enable discussion in meetings among superiors, subordinates and colleagues (iMCS1)
… To promote the continuous discussion regarding results, relevant data and action plans (iMCS2)
… To provide a common vision for the organization (iMCS3)
… To develop a common vocabulary within the organization (iMCS4)
… To enable you to focus on issues that affect the entire organization (iMCS5)
… To coordinate the different units in the organization with one another (iMCS6)
… To monitor results (dMCS1)
… To control the degree to which the objectives are achieved (dMCS2)
… To compare actual results with expected results (dMCS3)
… To review key indicators (dMCS4)
Belief (Be) and boundary (Bo) systems
To what extent do the following statements describe your organization? (1: strongly disagree and 5: strongly agree)
… The organization's mission statement clearly communicates the organization's core values to employees (Be1)
… Senior management communicates core values to employees (Be2)
… Employees are aware of the organization's core values (Be3)
… The organization's mission inspires employees (Br4)
… Our organization has a code of conduct to define appropriate behaviour on the part of employees (Bo1)
… The code of conduct informs our employees of behaviours that are impermissible (Bo2)
… Our organization has a system that informs employees of risks that must be avoided (Bo3)
… Employees know the organization's code of conduct (Bo4)

Interactive (iMCS) and diagnostic (dMCS) use of management control systems

Digital dynamic capabilities [The final composition of the items measuring DDCs differs from the original Warner and Wäger (2019) source. This is due to the validation process of the scale, which included both exploratory factor analysis (EFA) and confirmatory factor analysis (CFA). During this process, some items were excluded as they did not exhibit adequate psychometric properties. As a result, the scale was refined, leading to a final version that is more statistically robust and better suited to the context of this study.]

Indicate your level of agreement with the following statements regarding digital transformation (1: strongly disagree and 5: strongly agree). My organization…

Digital sensing

… Scans for technological trends (Se1)

… Screens of digital competitors (Se2)

… Senses customer-centric trends (Se3)

… Analyses scouted signals (Se4)

… Interprets future digital scenarios (Se5)

… Formulates digital strategies (Se6)

… Establishes a long-term digital vision (Se7)

… Enables an entrepreneurial mindset (Se8)

… Promotes a digital mindset (Se9)

Digital seizing

… Creates minimum viable products (Si1)

… Considers a lean start-up methodology (Si2)

… Uses a digital innovation lab (Si3)

… Balances internal and external options (Si4)

… Scales up innovative business models (Si5)

… Sets an appropriate speed of execution (Si6)

… Paces strategic responses (Si7)

… Rapidly reallocates resources (Si8)

… Accepts redirection and change (Si9)

Digital transforming

… Joins a digital ecosystem (Tr1)

… Interacts with multiple external partners (Tr2)

… Exploits new ecosystem capabilities (Tr3)

… Hires a chief digital officer (Tr4)

… Digitalizes business models (Tr5)

… Designs team-based structures (Tr6)

… Identifies digital workforce maturity (Tr7)

… Recruits digital talent externally (Tr8)

… Leverages digital knowledge inside the firm (Tr9)

Digital transformation success

In relation to the effect of digital transformation on your organization over the last three years, please indicate the extent to which you agree with the following statements (1: strongly disagree and 5: strongly agree):

… It is improving the competitive position in the environment in which your organization operates (Dts1)

… It is improving the image of the organization among different stakeholders (suppliers, clients, shareholders, workers, etc.) (Dts2)

… It is enhancing the value of the organization for its members, partners, shareholders and/or investors (Dts3)

… The expectations placed on your organization regarding the digital transformation are being met (Dts4)

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